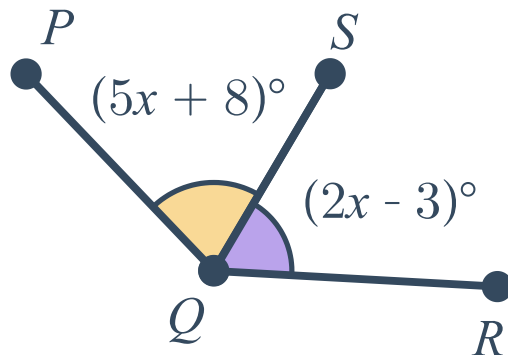


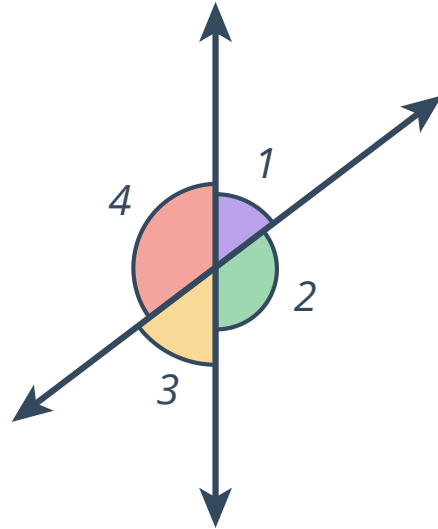
Proving angle relationships



- 1 The rays $\overrightarrow{DE}$ and $\overrightarrow{DF}$ are opposite rays.
Determine whether the following are true statements.
- a $\overrightarrow{DE}$ and $\overrightarrow{DF}$ are the legs of a straight angle.
 - b $m\angle EDF = 180^\circ$
 - c $\overrightarrow{DE}$ and $\overrightarrow{DF}$ are the legs of a right angle.
 - d $\overrightarrow{DE}$ and $\overrightarrow{DF}$ are not the legs of any angle.
 - e $m\angle EDF = 0^\circ$
- 2 The rays $\overrightarrow{DE}$ and $\overrightarrow{DF}$ are perpendicular to one another.
Determine whether the following statements are true.
- a $\angle EDF$ is a right angle.
 - b $m\angle EDF = 180^\circ$
 - c $m\angle EDF = 90^\circ$
 - d $\angle EDF$ is a straight angle.
- 3 Given that two angles form a linear pair, determine whether the following statements are always true.
- a They are complementary.
 - b They are congruent.
 - c They are adjacent.
 - d They are supplementary.
- 4 Consider the attached diagram.
Solve for x , given that $m\angle PQR = 145^\circ$.
Justify each steps using a two column proof.



- 5 Without directly using the vertical angles congruence theorem, construct a two column proof to prove that $\angle 1 \cong \angle 3$.

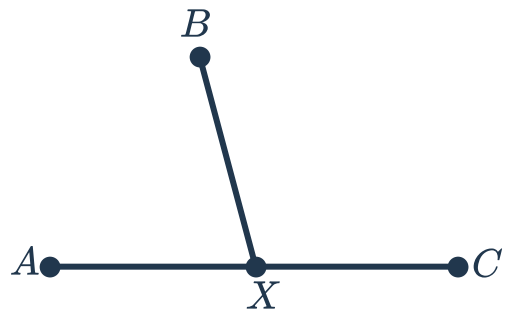


Additional questions

- 6 We are given that the two marked angles are congruent.
Construct a two column proof to solve for x .



- 7 Write a two column proof to prove the linear pairs postulate.
Given: $\angle AXB$ and $\angle BXC$ are a linear pair
Prove: $\angle AXB$ and $\angle BXC$ are supplementary



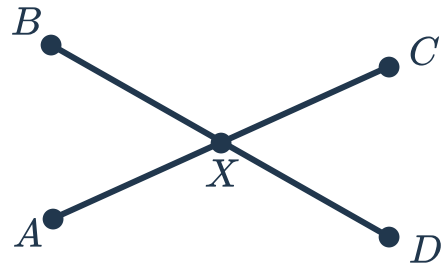
- 8 Write a two column proof to prove the vertical angles theorem.



Given:

- $\overline{AC}$ is a straight line segment
- $\overline{BD}$ is a straight line segment

Prove: $\angle AXB \cong \angle CXD$



- 9 Construct a two column proof to prove the congruent complements theorem.

Given:

- $\angle A$ and $\angle C$ are complementary
- $\angle B$ and $\angle C$ are complementary

Prove: $\angle A \cong \angle B$

- 10 Construct a two column proof for the theorem "all right angles are congruent".

Given: $\angle A$ and $\angle B$ are right angles

Prove: $\angle A \cong \angle B$

- 11 Write a two column proof for the theorem "if two angles are congruent and supplementary, both are right angles".

Given: $\angle A$ and $\angle B$ are congruent and supplementary

Prove: $\angle A$ and $\angle B$ are right angles